Calc 2 – Convergence Tests

Worksheet A

1. Use the integral test to determine whether $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ converges.

2. Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{2^n}$.

3. Use the root test on $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$.

4. Use limit comparison on $\sum_{n=1}^{\infty} \frac{1}{n^2 + n}$.

5. Determine convergence of $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$; is it absolute or conditional?

Calc 2 – Convergence Tests

Worksheet B

1. Use the integral test on $\displaystyle\sum_{n=1}^{\infty} \frac{n}{e^n}$.

2. Use the ratio test on $\displaystyle\sum_{n=1}^{\infty} \frac{2^n}{n!}$.

3. Use the root test on $\displaystyle\sum_{n=1}^{\infty} \left(\tfrac{2n+1}{3n}\right)^n$.

4. Use direct comparison: $\displaystyle\sum_{n=1}^{\infty} \frac{\sin^2 n}{n^2}$.

5. Determine convergence of $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1}$; is it absolute or conditional?

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Answer Key (Worksheets A & B)

Worksheet A.

1. $f(x) = 1/(x \ln x)$, with $u = \ln x$: $\int \frac{dx}{x \ln x} = \ln(\ln x)$, which $\to \infty$. **Diverges.**

2. $\frac{a_{n+1}}{a_n} = \frac{(n+1)!/(2^{n+1})}{n!/2^n} = \frac{n+1}{2} \to \infty > 1$. **Diverges.**

3. $\sqrt[n]{a_n} = \frac{n}{2n+1} \to 1/2 < 1$. **Converges.**

4. Compare with $b_n = 1/n^2$: $a_n/b_n = \frac{n^2}{n^2+n} \to 1$. Since $\sum 1/n^2$ converges, so does $\sum a_n$.

5. Alternating, $b_n = 1/n \downarrow 0$: converges by Leibniz. But $\sum 1/n$ diverges. **Conditionally convergent.**

Worksheet B.

1. $\int_1^\infty xe^{-x}\,dx = [-xe^{-x} - e^{-x}]_1^\infty = 2/e$. Finite, so the series **converges**.

2. $\frac{a_{n+1}}{a_n} = \frac{2}{n+1} \to 0 < 1$. **Converges.**

3. $\sqrt[n]{a_n} = \frac{2n+1}{3n} \to 2/3 < 1$. **Converges.**

4. $0 \le \frac{\sin^2 n}{n^2} \le \frac{1}{n^2}$, and $\sum 1/n^2$ converges. **Converges** by direct comparison.

5. Alternating; $b_n = \frac{n}{n^2+1}$ is decreasing for $n \ge 1$ and $\to 0$. So converges by Leibniz. $\sum \frac{n}{n^2+1}$ behaves like $\sum 1/n$ (limit comparison): diverges. **Conditionally convergent.**